

A. **Higher–Order Derivatives.** We may take repeated partial derivatives.

The second column forms the 1st order derivatives.

The third column forms the 2nd order derivatives.

$$f_{xx} = \frac{\partial^2 f}{\partial x^2} =$$

$$f_x = \frac{\partial f}{\partial x} =$$

$$f_{xy} = \frac{\partial^2 f}{\partial y \partial x} =$$

$$f = xy^2 + e^y$$

$$f_{yx} = \frac{\partial^2 f}{\partial x \partial y} =$$

$$f_y = \frac{\partial f}{\partial y} =$$

$$f_{yy} = \frac{\partial^2 f}{\partial y^2} =$$

Clairaut's Theorem. If f has continuous partial derivatives of all orders

Example 1. Find $f_{xyzzzyxz}(x, y, z)$ where $f(x, y, z) = \frac{z^2 e^{xy^2 + \sin(x+y)}}{x}$

Solution. We change the order of differentiation so

This gives

$$f_z(x, y, z) =$$

$$f_{zz}(x, y, z) =$$

$$f_{zzz}(x, y, z) =$$

The remaining derivatives will maintain the value:

B. **Chain Rule.** Consider the relation:

$$z = x^2 y$$

but suppose x and y further depend on variables s and t .

$$\begin{cases} x = st \\ y = e^{st} \end{cases}$$

The equation of the tangent plane reads

$$\Delta z =$$

If we divide by Δs then take a limit as $\Delta s \rightarrow 0$ we obtain

We call the first expression for $(\partial z)/(\partial s)$ the **chain rule**.

There is another way to obtain it. First create a tree of dependencies.

Next identify all paths falling from z to s .

Convert each path to a product of two partial derivatives, then add.

$$\frac{\partial z}{\partial s} =$$

We can also apply this process, using paths falling from z to t , to find

$$\frac{\partial z}{\partial t} =$$

Example 2. Let $z = f(x, y)$ be a differentiable function and let

$$\begin{cases} x = s + t \\ y = s - t \end{cases}$$

Show that: $\frac{\partial z}{\partial s} \frac{\partial z}{\partial t} = \left(\frac{\partial z}{\partial x} \right)^2 - \left(\frac{\partial z}{\partial y} \right)^2$.

Solution. First we construct a tree of dependencies.

In that tree, we mark the partial derivative corresponding to each edge.

Next we execute the chain rule to compute

$$\frac{\partial z}{\partial s} =$$

$$\frac{\partial z}{\partial t} =$$

Lastly we multiply the results to obtain

$$\frac{\partial z}{\partial s} \frac{\partial z}{\partial t} =$$

C. Chain Rule and Gradient. We relate the chain rule to paths. Consider

$$f(x, y, z) = xe^{y/z}$$

where x, y, z further depend on variable t .

$$\begin{cases} x = t^2 \\ y = -t \\ z = 2t \end{cases}$$

These dependencies define a path.

$$\langle x, y, z \rangle = \vec{x} = \vec{r}(t) =$$

We construct a tree dependencies and label edges.

By the chain rule, but now written as a dot product

$$\frac{df}{dt} =$$

This first vector function is called the **gradient**:

The second vector function is none other than:

Chain Rule: Paths. If f depends differentiably on differentiable $\vec{x} = \vec{r}(t)$

$$\frac{df}{dt} =$$

Let's apply this to our setting, where $f(x, y, z) = xe^{y/z}$ and $\vec{r}(t) = \langle t^2, -t, 2t \rangle$.

$$\vec{\nabla} f(x, y, z) =$$

$$\vec{r}'(t) =$$

$$\frac{df}{dt} =$$